RPHB Secondary Data Power

All replication analysts are asked to complete three tasks:

- Provide a best estimate of the focal effect size, expressed as a partial correlation.
- Answer four questions determining whether the replication analysis is simple or complex and an additional two questions on the validity of the replication dataset. If the analysis is complex, escalate to the study lead to ensure consistency across projects.
- Compute two power estimates based on the secondary data's sample size: the power to detect 75% of the original effect size and the power to detect 100% of the effect.

Partial correlations

We recommend using the effectsize R package with the reported or calculated test statistic and the effective sample size, when relevant. Ignore the sign of the test statistic – it's not relevant to the power computations. The following scenarios should cover many cases. For cases not covered here, either identify or adapt the closest defensible conversion or escalate to the scientific lead.

Original reporting	Conversion function
t-statistic + df	t_to_r(t, df_error)
F-statistic (1 df) + df	F_to_r(f, df = 1, df_error)
χ^2 (1 df, i.e. a 2×2 table) + N	chisq_to_phi(chisq, n, nrow = 2, ncol = 2)
Linear coefficient + SE	t_to_r(t = coef/SE, df_error)
Logistic coefficient + SE	z_to_r(z = coef/SE, n)

Replication design questions

Please answer the following four questions. If any are answered yes, please escalate this case to the scientific lead to ensure consistency in approach.

- Are observations nested in groups (people within schools/clinics/countries, repeated measures within person) such that you'd use a multilevel/mixed model?
- Does the dataset have a complex sample design — weights, strata, clusters — that your analysis needs to account for?
- Is the effect a mediation or structural-equation-model (SEM) result, rather than a single model coefficient?
- Is the outcome a rare event or count variable?

Additionally, please answer two questions about the validity of the replication dataset:

- Are there covariates in the original analysis that are unavailable in the replication data? Will this affect your ability to conduct a good faith, high-quality replication? Why or why not?

- Are the replication cases/observations fully independent of the original study? Conversely and equivalently, are there any individuals or units who appear in both the original study and the replication data?

Power estimates

For standard replication designs, please compute the Tier 1 and Tier 2 power estimates. Recommended code is below. For non-standard designs, please escalate to the scientific lead.

```
library(pwr)

r_original <- ___ # your best-estimate partial r for the original effect
N          <- ___ # replication dataset's usable N
k          <- ___  # total number of predictors in your replication model

alpha <- 0.05
v <- N - k - 1

r_tier1 <- 0.75 * r_original
r_tier2 <- 1.00 * r_original

f2_tier1 <- r_tier1^2 / (1 - r_tier1^2)
f2_tier2 <- r_tier2^2 / (1 - r_tier2^2)

power_tier1 <- pwr.f2.test(u = 1, v = v, f2 = f2_tier1, sig.level =
alpha)$power
power_tier2 <- pwr.f2.test(u = 1, v = v, f2 = f2_tier2, sig.level =
alpha)$power

cat(sprintf("Tier 1 power (75%% of effect, target 90%%): %.1f%%\n", power_tier1
* 100))
cat(sprintf("Tier 2 power (full effect, target 80%%):    %.1f%%\n", power_tier2
* 100))
```